INVESTIGATING THE STABILITY OF SPINNING WORMHOLES WITH NONLINEAR EVOLUTIONS

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We investigate the stability of spinning Ellis–Bronnikov wormholes using fully nonlinear 3+1 numerical relativity simulations of the drainhole family, whose static member we evolved in our previous work. We construct the stationary metrics with a spectral solver that is exact to $\sim 10^{-10}$ at slow spin and $\sim 10^{-7}$ near the extremal limit, and validate it against the published symmetric family of Kleihaus and Kunz. In doing so we correct an inconsistency in a published field equation and map the drainhole family, for which no fast-spin solutions were known, up to $v_e = 0.99$; the 3D initial data converge at fourth order. We find that rotation slows the radial instability: on exact backgrounds the slowly spinning drainhole collapses with an e-fold time of 6.8 against the static 5.8, and at rotation velocity $v_e = 0.51$ the throat radius falls by only 0.6% in the time the static throat takes to form a horizon. Second-order perturbative initial data instead seed the instability at the level of their own truncation error.

I. INTRODUCTION

The Ellis–Bronnikov wormhole, a throat held open by a phantom scalar field [1–4], is unstable: depending on the sign of the perturbation it collapses to a black hole or inflates, on a timescale set by the throat radius [5–9]. Nonlinear 3+1 evolutions confirm both branches for the lone throat [10] and carry them into binaries, whose every channel ends in collapse, inflation or a merged horizon within a few dynamical times [11]. If traversable wormholes are to persist anywhere, something must shut this instability off.

Rotation is the oldest candidate [12]. Spinning Ellis–Bronnikov solutions are known perturbatively [13] and, beyond perturbation theory, from the elliptic solver of Kleihaus and Kunz [14, 15]: a family that starts at the static throat and ends, at equatorial rotation velocity $v_e = 1$, on an extremal Kerr black hole, the phantom field vanishing in the limit; no closed form exists [16]. On the stability side, Azad, Blázquez-Salcedo, Khoo and Kunz [17, 18] carried the radial mode to second order in spin: the static unstable mode weakens, with a positive $O(J^2)$ correction for every asymmetry, while for nearly symmetric wormholes a second unstable mode grows out of a static zero mode; the two cross near 30% of the limiting angular momentum, where the quadratic approximation gives out, and the authors conjecture the modes merge and the instability disappears, as purely imaginary modes do in five dimensions [19]. The charged analogue warns against optimism: there the merged modes continue as a complex pair, growth that oscillates, and the growth rate reaches zero only in the extremal limit [20]. Which continuation rotation takes in four dimensions is exactly the question a perturbative method cannot answer, and no rotating wormhole has ever been evolved nonlinearly.

This paper builds the tools for that evolution and takes its first step. We construct two families of backgrounds as exact stationary solutions. The symmetric family ($\gamma = 0$), massless when static, is the one with published data—the curves of Ref. [14] and the scalar charges of Ref. [15]—and we use it to validate the solver and the 3D initial data across the spin range. The drainhole family ($\gamma = -\pi$), whose static member is the mass-1 drainhole evolved in Refs. [10, 11], has no published fast-spin data at all; it is the family we evolve. The solver is gated on identities it never imposes, a discipline that pays immediately: one of the published field equations is inconsistent (Sec. II A), and second-order perturbative initial data turn out to seed the instability at their own error level (Sec. V). We then put the backgrounds on the AMR grid, verify them at fourth order, and evolve the drainhole along a ladder of five spins (Table II), starting from its slow-spin end. The result so far is one-sided but clean: rotation slows the collapse, as the perturbative sign predicts.

Section II gives the family and the corrected field equations; Sec. III the solver, the validation against the published data, and the map of the parameter space (Table I); Sec. IV the 3D initial data and the evolution ladder (Table II); Sec. V the slow-spin evolutions; Sec. VI the conclusions. Appendix A holds solver details and the comparison with the anchor solutions of Ref. [15]. We use $G = c = 1$. Table I quotes the backgrounds in units of the throat parameter η_0; the evolutions use the code units of Refs. [10, 11], $\eta_0 = 2$, in which the static drainhole has mass $M = 1$.

II. THE SPINNING WORMHOLE FAMILY

The matter model is that of our previous work [10, 11]: Einstein gravity coupled to a phantom (ghost) scalar field

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ψ,

$$S = \frac{1}{16\pi}\int d^4x\,\sqrt{-g}\,(R + 2\,g^{\mu\nu}\partial_\mu\psi\,\partial_\nu\psi)\,, \qquad (1)$$

whose kinetic term has the reversed sign that makes ψ a ghost. Varying the action with respect to the metric and to ψ gives the field equations of the model [Eqs. (2)–(3) of Ref. [15]],

$$R_{\mu\nu} = -2\,\partial_\mu\psi\,\partial_\nu\psi, \qquad \partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\psi\big) = 0\,: \qquad (2)$$

Einstein's equations with the phantom's stress tensor as source, and the wave equation for the phantom on the curved background. Those papers evolved its best-known static wormhole—the Ellis–Bronnikov drainhole with throat parameter $a = 2$ and end masses ±1—and measured the instability that this paper asks rotation to cure. Spinning that wormhole up requires the stationary rotating solution of the same theory, and none is known in closed form [16]: it has to be constructed numerically, inside an *ansatz*—an assumed form of the metric with stationarity and axisymmetry built in and a few unknown functions left for the field equations to determine. We take the ansatz of Refs. [14, 15], who first constructed these solutions,

$$\begin{aligned} ds^2 = -e^f dt^2 + p^2 e^{-f}\Big[e^\nu\left(d\eta^2 + h\,d\theta^2\right) \\ + h\sin^2\theta\,(d\varphi - \omega\,dt)^2\Big], \end{aligned} \qquad (3)$$

with $\eta \in (-\infty, \infty)$ and four unknown functions f, p, ν, ω of η and θ [Eq. (4) of Ref. [15]]. The field equation for p decouples from the others, and its solution with asymptotically flat boundary conditions is $p = 1$ [Eqs. (5)–(6) of Ref. [15]]; we set $p = 1$ from here on. The fixed function $h = \eta^2 + \eta_0^2$ is what makes this a wormhole ansatz. In the trivial limit $f = \nu = \omega = 0$ the spatial metric is $d\eta^2 + h\left(d\theta^2 + \sin^2\theta\,d\varphi^2\right)$, so $\sqrt{h}$ is the radius of the sphere at coordinate η—and by construction it never vanishes: it falls from η^2 at large $|\eta|$, bottoms out at η_0 at $\eta = 0$, and rises again. The spacetime therefore has no center; a minimal sphere of radius set by η_0, the throat, joins the two asymptotically flat ends $\eta \to \pm\infty$. Choosing h hard-wires this two-ended, horizonless topology into Eq. (3), while f, ν, ω deform the geometry on top of it (h also recurs as the weight in every equation below); conversely, $\eta_0 \to 0$ closes the throat and lets the same ansatz hold a black hole, which is how extremal Kerr enters in Sec. II A. Our previous wormhole sits inside this form: with $\omega = \nu = 0$ and the closed-form arctan profile for f [15], Eq. (3) is the static drainhole of Refs. [10, 11]—η is the isotropic radius r used there, unfolded through the throat by $\eta = r - \eta_0^2/4r$ (Sec. IV), and $\eta_0 = a$ its throat parameter—while $f = 0$ gives the massless Ellis wormhole. Everything the ansatz (3) leaves open—the metric functions $f(\eta,\theta)$ (the redshift of the two ends), $\omega(\eta,\theta)$ (the frame dragging) and $\nu(\eta,\theta)$, together with the phantom field—must come from the field equations (2): determining them at each spin is what Sec. III is for. Substituting the ansatz (3) into the field equations (2) reduces them in three steps. We follow the reduction of Kleihaus and Kunz [14], given in full in Sec. II of Ref. [15]; equation numbers in brackets below refer to Ref. [15]. First, the phantom field equation—the wave equation in Eq. (2), which stationarity and axisymmetry reduce to a function of η alone—integrates in closed form to $\psi = D\,[\arctan(\eta/\eta_0) - \pi/2]$ [Eqs. (7)–(9)]: the scalar enters the rest of the problem only through its constant of integration D, the scalar charge. Second, because $\partial_\mu\psi$ points along η, the source term $-2\,\partial_\mu\psi\,\partial_\nu\psi$ survives only in the $\eta\eta$ Einstein equation; each other nontrivial component (tt, $t\varphi$, $\theta\theta$, $\varphi\varphi$, $\eta\theta$) is a statement about the metric functions alone, and these organize by order and role. Two are second-order elliptic equations,

$$\begin{aligned} 0 &= \partial_\eta(hs\,\partial_\eta f) + \partial_\theta(s\,\partial_\theta f) \\ &\quad - hs^3 e^{-2f}\left[h(\partial_\eta\omega)^2 + (\partial_\theta\omega)^2\right], \qquad (4) \\ 0 &= \partial_\eta\left(h^2 s^3 e^{-2f}\partial_\eta\omega\right) \\ &\quad + \partial_\theta\left(hs^3 e^{-2f}\partial_\theta\omega\right), \qquad (5) \end{aligned}$$

with $s \equiv \sin\theta$ [Eqs. (10)–(11) of Ref. [15]]: Eq. (4) is a Laplace-type equation for the redshift f, sourced by the squared gradients of the frame dragging—rotation gravitates—and Eq. (5) is a total divergence, the local form of angular-momentum conservation, which integrated over the domain forces the same J through every $\eta = $ const surface. The single off-diagonal component is only first order in ν,

$$\begin{aligned} 0 = -hs\,\partial_\eta f\,\partial_\theta f + h\cos\theta\,\partial_\eta\nu \\ + \eta s\,\partial_\theta\nu + h^2 s^3 e^{-2f}\partial_\eta\omega\,\partial_\theta\omega, \end{aligned} \qquad (6)$$

so ν follows by integration once f and ω are known [Eq. (13)]. (The remaining diagonal components give a second-order equation for ν as well—the published Eq. (12), shown misprinted in Sec. II A; we never use it.) Third, the one sourced component, $R_{\eta\eta} = -2D^2/h^2$ [Eq. (18)], rearranges into an algebraic identity for the charge [Eq. (19)]: the metric functions fix D^2 at every point, so its constancy over the domain is a solution test that comes for free. Three unknown functions thus face more equations than they need; the Bianchi identities make the overdetermined set consistent on a true solution, and a numerical scheme must choose which subset to impose—a choice Sec. II A shows matters. The boundary conditions select the family: Minkowski at $\eta \to +\infty$, and $f \to \gamma$, $\omega \to \omega_\infty$ at $\eta \to -\infty$, so the constant γ measures how unequal the two ends are and ω_∞ sets the spin.

A solution of this system is three functions and one number, $(f, \nu, \omega; D)$. Every physical quantity we quote—the columns of Table I—is read off them directly [14, 15]:

- far from the throat the metric settles to the field of a point mass, $f \to -2\mu_+/\eta$ as $\eta \to +\infty$ and $f \to \gamma + 2\mu_-/\eta$ as $\eta \to -\infty$, so the end masses $\mu_\pm$ are the $1/\eta$ coefficients of f;

- likewise $\omega \to 2J/\eta^3$, and the conservation law (5) forces the same J on both ends; the charges obey the Smarr-type relation $\mu_+ + \mu_- = 2\,\omega_\infty J$ [15], which the solver never imposes and uses as an acceptance gate;

- the throat is the sphere of minimal area, at $\eta_t = 0$ when the two ends are equal ($\gamma = 0$) and displaced toward the positive-mass end otherwise ($\eta_t = \eta_0/2$ for the static drainhole): its equatorial circumference gives the radius $R_e = e^{-f/2}\sqrt{h}\,\big|_{\eta_t,\,\theta=\pi/2}$, its meridians the polar radius R_p, its area the areal radius R_A—three numbers that coincide while the throat is round and split as rotation flattens it;

- the throat is dragged at the angular velocity $\Omega = \omega\,\big|_{\eta_t,\,\theta=\pi/2}$, and $v_e = R_e\,\Omega$ is its equatorial linear velocity in units of c: the dimensionless spin label used throughout, 0 when static and 1 at the extremal Kerr endpoint.

The same read-offs, applied to the 3D grid data instead of the solver's functions, are how Sec. IV verifies the initial data against Table I.

We construct two one-parameter families, labelled by ω_∞ (equivalently J or v_e):

(a) *the drainhole family*, $\eta_0 = 2$, $\gamma = -\pi$: its static member is the Ellis drainhole with throat parameter $a = 2$ and unequal end masses $\mu_+ = 1$, $\mu_- = -1$, the wormhole of Refs. [10, 11] (the asymmetry $C = 1$ of Refs. [17, 18]). This is the family we evolve.

(b) *the symmetric family*, $\gamma = 0$ ($C = 0$): equal masses on both ends, massless and identical to the Ellis wormhole when static. This is the family of Refs. [14, 15], the only one with published solutions, and we use it to validate the solver (Sec. III) and the 3D initial data (Sec. IV); for the 3D checks it is rescaled $\eta_0 = 1 \to 2$, so its throat sits on family (a)'s grid.

A. An inconsistency in the published field equations

Refs. [14, 15] close the system with a second-order equation for ν, Eq. (12) of Ref. [15]. That equation is inconsistent with the rest. Extremal Kerr fits the ansatz (3) in closed form (with $\eta_0 = 0$), and substituted symbolically, at 160-digit arithmetic, it satisfies Eqs. (4)–(6) and the D^2 identity exactly—while leaving an $O(10^{-2})$ residue in the printed ν equation alone. The error also shows dynamically: solutions of the printed elliptic system converge at every resolution, yet carry a violation of the unimposed Eq. (6) and D^2 identity that scales as J^4 and does not shrink with the grid—the fingerprint of a wrong continuous equation, not of numerics (first seen as a spread of 5.5×10^{-7} in D^2 at $J = 0.25$). We therefore never solve the printed ν equation: ν and D^2 are obtained jointly from Eq. (6) and the D^2 identity, and the printed equation is kept only as a reported diagnostic. Every published conclusion of Refs. [14, 15] that we can test survives the correction (Sec. III, Appendix A); the published positive-asymmetry scalar charges do not (Appendix A).

III. BACKGROUNDS ACROSS THE SPIN RANGE

The solver turns the system of Sec. II into numbers in five steps (Fig. 1); Appendix A gives the details. First, the infinite domain is made finite: the coordinate $x = (2/\pi)\arctan(\eta/\eta_0)$ maps both asymptotic ends to $x = \pm 1$, and the unknowns f and ω are expanded in Chebyshev polynomials of x and even cosine modes of θ. The expansion is spectral: for smooth solutions the error falls exponentially with the number of modes, so modest grids reach near machine precision. Second, the two elliptic equations (4)–(5) are solved for the expansion coefficients by Newton iteration—guess, linearize, correct—until their residual sits at round-off. Third, ν and the scalar charge are recovered without ever touching the misprinted second-order ν equation: the first-order constraint (6) and the D^2 identity together overdetermine ν and the single number D^2, so both are extracted as one linear least-squares problem—and the redundancy is a feature, since the leftover misfit measures the error. Fourth, each family is walked in spin: every converged solution seeds the Newton iteration at a slightly higher spin, with J as the continuation parameter (ω_∞ is not monotone along a family; Appendix A), and the grid refines automatically, from 48×12 up to 240×60, whenever the acceptance gates fail. Fifth—the point of the whole design—a solution is accepted only on identities the solver never imposes: the spread of D^2 over the domain ($\leq 10^{-11}$), the Smarr relation ($\leq 10^{-10}$), equality of the angular-momentum flux through the two ends ($\leq 1.3 \times 10^{-9}$), the asymptotic tails ($\leq 5 \times 10^{-12}$), and an independent, symbolically generated residual of the full Einstein equations, evaluated out to 20 throat radii: 1.0×10^{-10} at $J = 0.25$ and 4.2×10^{-10} at $J = 1$ against the static member's 2×10^{-11} round-off floor—where the $O(J^2)$ perturbative data give 3×10^{-5} and 1×10^{-2} on the same meshes—staying near 10^{-9} up to $v_e \approx 0.85$ and reaching $\sim 10^{-7}$ at $v_e = 0.91$–0.99. The high-spin 10^{-7} is round-off, not solution error: it does not shrink on finer grids, while the mass and scalar charge come out identical to 12 digits across them; and the test does catch genuine failures, scoring a deliberately under-resolved solution at 10^{-4}, which the resolution check rejects. Beyond 20 throat radii the residual's normalization inflates round-off and the test is switched off; that outer region is covered by the resolution study, by the end charges, and by the 3D constraint checks of Sec. IV. The prod-

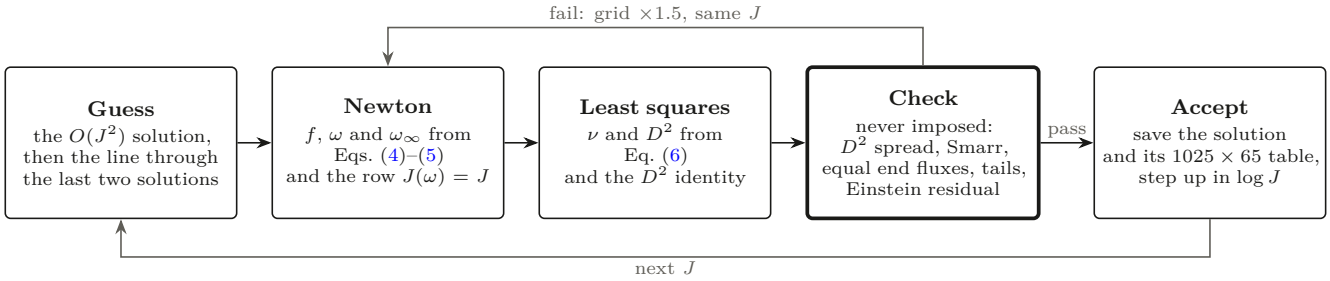


FIG. 1. The solver loop for one family (Sec. III). Thin boxes impose equations; the heavy box tests identities the solver never imposes, and only it decides acceptance. A failed check refines the Chebyshev grid (48×12 up to 240×60) and solves the same J again; an accepted solution seeds the next J. The loop gave 462 accepted solutions over 20 continuation legs, both families to $v_e \approx 0.99$.

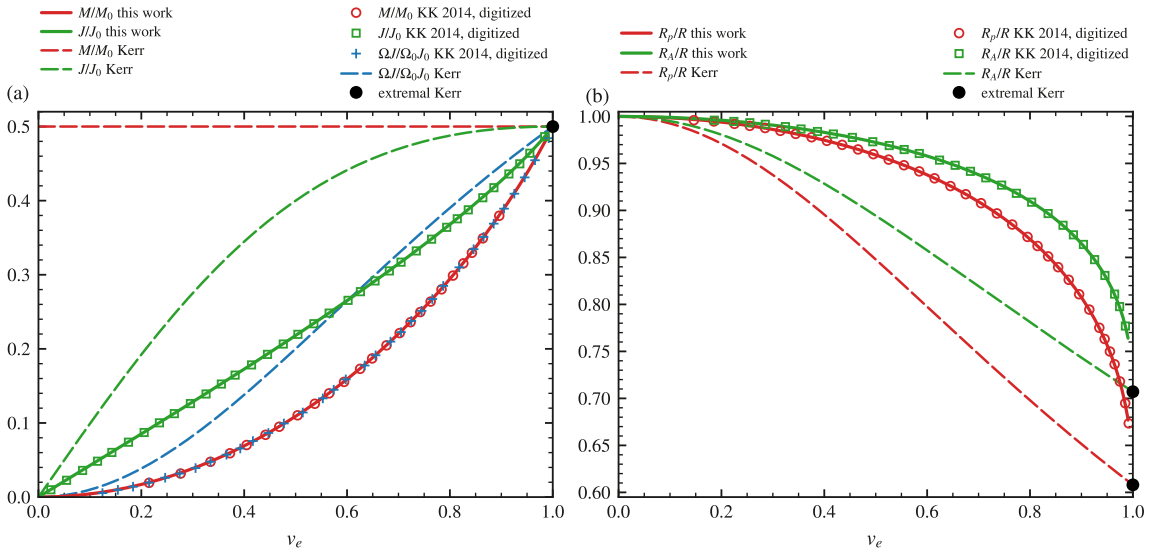


FIG. 2. The symmetric family ($\eta_0 = 1$, $\gamma = 0$; 81 solutions, $v_e = 0.02$–0.991, plus the exact static member) in the normalization of Ref. [14] ($M_0 = R_e$, $J_0 = R_e^2/2$, $\Omega_0 = 1/R_e$): our curves, the digitized points of their Fig. 1 (open symbols), and closed-form Kerr (dashed). The digitization's reading error, measured by reading their printed Kerr curves the same way, is rms 1–6×10^{-4}; digitized minus ours is rms 2.5–8×10^{-4}, max 2.5×10^{-3}, over all five curves. The two solvers agree to the precision a printed figure carries.

ucts: 462 converged solutions over 20 continuation legs, global charges stable to 10 digits across three grids, and, for evolution, each background exported as a 1025×65 table whose bilinear interpolation error is $\le 10^{-7}$ up to $v_e = 0.9$.

Figure 2 is the validation: in the normalization of Ref. [14], our symmetric family lands on the digitized curves of their Fig. 1 to the digitization's own reading error. Their four anchor solutions at $\omega_\infty = 0.5$ [15] split: the $\gamma = 0$ and $\gamma = -1$ scalar charges agree to all printed digits, while $\gamma = 1$ and $\gamma = 2$ differ by 2×10^{-4} and 6.6×10^{-3} from our grid-converged values—their finite-difference error, growing with γ (Appendix A).

Figure 3 and Table I are the map of the parameter space the foundation papers left open. Published data exist only for the symmetric family and only as figures; the massive drainhole family—the one whose static member has been evolved, and the one from which binaries can be built [11]—is mapped here from slow spin to $v_e = 0.99$. Two of its features matter later. First, both families violate the Kerr bound: the symmetric family everywhere ($J/\mu_+^2 \approx 85$ already at $v_e = 0.24$, falling monotonically to 1 at extremality, since its static member is massless), the drainhole family by up to $\sim 20\%$ at intermediate spin. A collapsing overspun throat cannot settle to a Kerr black hole of the same mass and spin, which makes these backgrounds sharp cosmic-censorship initial data. Second, the drainhole's negative far-end mass μ_- rises through zero at $v_e \approx 0.7$ and turns positive: fast rotation un-does the static drainhole's negative-mass end.

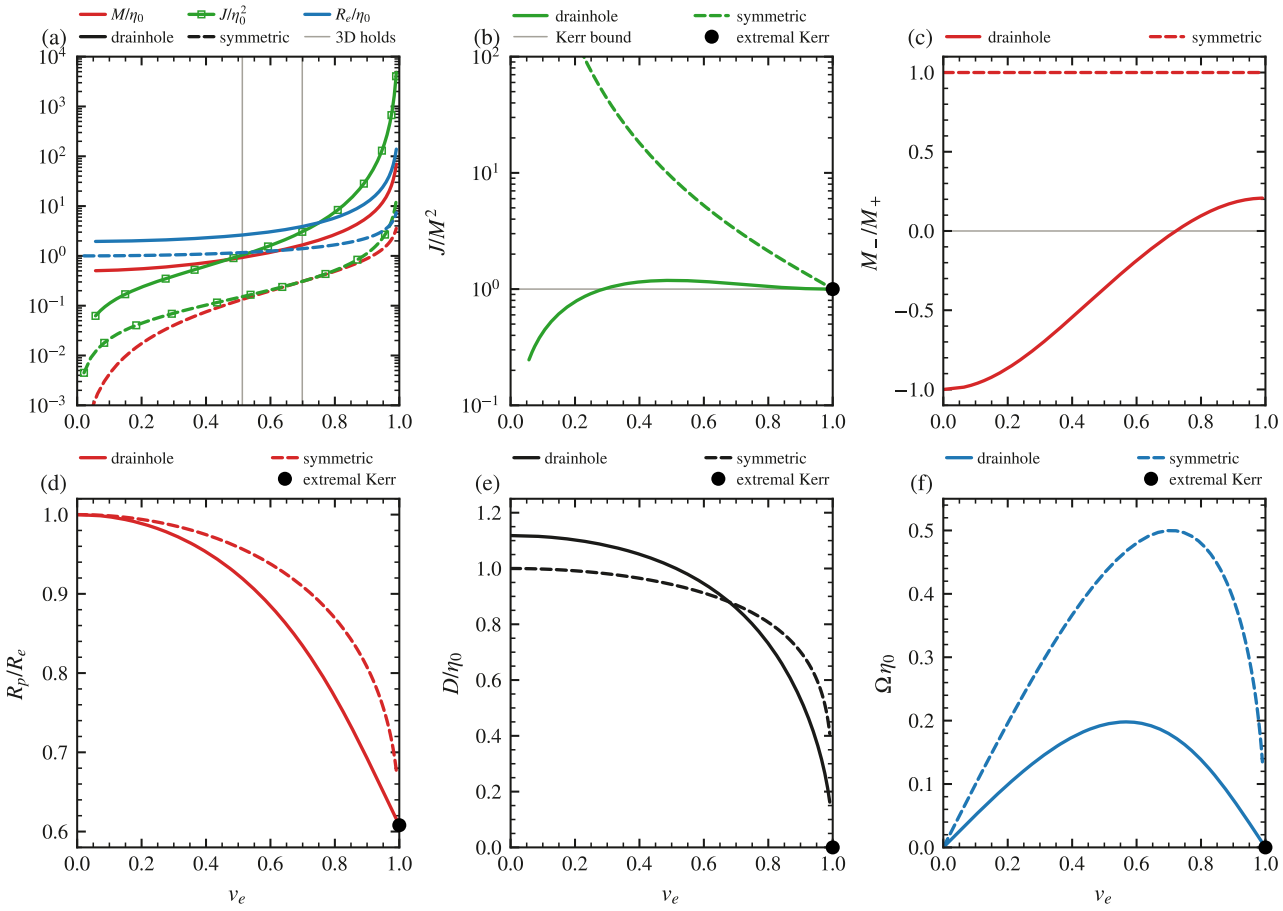


FIG. 3. The two families across the spin range (solid: drainhole, $\eta_0 = 2$, $\gamma = -\pi$; dashed: symmetric, $\eta_0 = 1$, $\gamma = 0$), against v_e: (a) mass, angular momentum and equatorial throat radius; (b) J/M^2 against the Kerr bound; (c) the ratio of end masses; (d) the flattening R_p/R_e; (e) the scalar charge; (f) the throat angular velocity, which peaks and falls to zero at extremality. Grey verticals in (a): the spins of the first two drainhole holds. The symmetric family violates the Kerr bound at every spin and the drainhole family exceeds it by up to $\sim 20\%$ at intermediate spin; both families end on extremal Kerr.

IV. 3D INITIAL DATA AND EVOLUTION SETUP

The stationary metric (3) goes onto the Cartesian AMR grid through the isotropic radius r, $\eta = r - \eta_0^2/4r$: the far end $\eta \to -\infty$ compactifies to the puncture $r = 0$, as for the static drainhole [10]. The induced metric and the fields are read from the tables; the lapse and shift start from the stationary data, $\alpha = e^{f/2}$, $\beta^\varphi = -\omega$, so the extrinsic curvature $K_{ij} = (D_i\beta_j + D_j\beta_i)/2\alpha$ has only $r\varphi$ and $\theta\varphi$ components and the scalar momentum vanishes; at the puncture the frame rotates at ω_∞. Evolution uses `GRTeclyn` [26–28] with the setup of Refs. [10, 11]: CCZ4 [23], 1+log slicing [24], Gamma-driver shift [25], fourth-order stencils, RK4, Kreiss–Oliger dissipation, $\Delta t = 0.02\,\Delta x$, and a floored conformal factor at the puncture.

Speeding up the runs. The timestep of Refs. [10, 11], $\Delta t = 0.02\,\Delta x$, sits far below both the advective CFL limit and the Kreiss–Oliger stability bound ($\Delta t \lesssim 0.23\,\Delta x$ at dissipation strength $\sigma = 0.1$); it was forced empirically, by evolutions that failed at larger steps for a then-unknown reason. We traced the cause on the static hold: a grid-scale checkerboard mode pinned to the fixed coarse–fine interfaces of the nested refinement boxes, dissipation-starved at $\sigma = 0.1$, whose growth rate rises with the timestep—it breaks out of the smooth collapse background at $t \approx 28$ for $\Delta t = 0.05\,\Delta x$ and $t \approx 43$ for $0.035\,\Delta x$, while at $0.02\,\Delta x$ it survives as a slow exponential growth (e-fold time $\sim$9) of the Hamiltonian constraint on the refinement level that carries the box edges. Constraint norms evaluated per refinement level, with cells covered by finer levels masked out, expose the mode by $t \approx 10$; the domain-wide norm, dominated by the coarse grid, is blind to it until breakout. The same slow variant drives the late-time constraint growth ($t \gtrsim 60$) that helped limit the trusted windows of the static and binary evolutions of Refs. [10, 11]. The cure is to strengthen the dissipation where the mode lives: at $\sigma = 0.3$ the evolution with $\Delta t = 0.05\,\Delta x$ is clean through the full test

TABLE I. The two families at representative spins, in units of η_0 ($G = c = 1$): rotation velocity v_e, angular momentum, mass of the $\eta \to +\infty$ end, ratio of end masses, scalar charge, equatorial throat radius, flattening, and throat angular velocity—each read off the solution's functions as defined in Sec. II. Every row is a converged solution that passed the acceptance gates of Sec. III. The published record covers family (b) only, as curves [14, 15], and we use it for validation; family (a), the one we evolve, is mapped here for the first time.

family	v_e	J/η_0^2	μ_+/η_0	μ_-/μ_+	D/η_0	R_e/η_0	R_p/R_e	$\Omega\,\eta_0$
(a) drainhole	0.213	0.2535	0.5574	−0.850	1.1004	2.0319	0.9873	0.1046
	0.513	1.0278	0.9302	−0.339	1.0007	2.6271	0.9194	0.1952
	0.699	3.0417	1.6557	−0.032	0.8598	3.8864	0.8351	0.1798
	0.900	34.550	5.8400	+0.179	0.5327	11.911	0.6926	0.0755
	0.950	152.16	12.316	+0.201	0.3785	24.751	0.6504	0.0384
	0.990	4053.8	63.666	+0.208	0.1698	127.36	0.6165	0.0078
(b) symmetric	0.239	0.0542	0.0252	1	0.9881	1.0299	0.9913	0.2323
	0.434	0.1158	0.0906	1	0.9583	1.1101	0.9699	0.3911
	0.667	0.2683	0.2666	1	0.8854	1.3420	0.9197	0.4969
	0.847	0.7007	0.6305	1	0.7653	1.8836	0.8440	0.4499
	0.898	1.0989	0.8685	1	0.7043	2.2722	0.8092	0.3952
	0.952	2.4701	1.4383	1	0.5971	3.2703	0.7539	0.2911
	0.990	12.235	3.4312	1	0.4157	7.0595	0.6795	0.1402

window, with interface-level constraint norms *below* the $\sigma = 0.1$, $\Delta t = 0.02\,\Delta x$ reference, and at the production timestep $\sigma = 0.4$ removes the slow growth entirely, leaving the interface-level constraint norms 10^2–10^4 times below the $\sigma = 0.1$ reference by $t = 100$; $\sigma = 0.2$ only postpones the breakout, setting the floor at $\sigma = 0.3$. The larger timestep does not transfer to the spinning backgrounds: on the $L = 128$ production grid at $v_e = 0.51$, $\Delta t = 0.05\,\Delta x$ fails by $t \approx 48$ (the conformal factor collapses to its floor and K runs away) with either dissipation strength, while the same data at $\Delta t = 0.02\,\Delta x$ runs clean to $t = 100$. The production triplets therefore keep $\Delta t = 0.02\,\Delta x$ and adopt the raised dissipation alone; its one measured side effect is a constant offset of the collapse clock (the stronger initial smoothing advances the static hold's collapse by ∼4 time units), so rates are always compared at matched σ.

Preparing the long runs. Evolutions that must outlive the outer-boundary crossing time $t \approx L$ need the boundary made harmless. Kreiss–Oliger dissipation cannot do it—its strength falls as the sixth power of wavelength, so the sponge shell of Refs. [10, 11] is transparent to the long outgoing gauge pulse—and the Sommerfeld condition assumes flat-space asymptotics the wormhole does not have (mass ∼2 sits inside the box). We therefore add a damping layer in the same shell: every evolved field is driven back to the stationary background at rate A, $\partial_t u \to \partial_t u - A\,f(r)\,(u - u_{\rm bg})$, with $f(r)$ the sponge's quartic ramp and $u_{\rm bg}$ the run's own initial data, rebuilt exactly from the background tables; $A = 0$ recovers the base scheme identically. Validated on the static hold at $A = 1$ against the reference evolution: $|K|$ falls two orders of magnitude across the shell, while everything inside it—the throat radii to four digits, the constraint norms, the near-zone ring—is unchanged, so the layer absorbs at all wavelengths without touching the interior. The same comparison shows the late-time growth previously attributed to boundary reflection is unchanged by the absorber: it is sourced by the collapse region itself, and is removed by the raised dissipation above. Wave-extraction spheres sit inside the damping layer, where the signal is untouched.

Stability claims follow the triplet protocol of Ref. [22]: every quoted growth must have an e-fold time that converges to a nonzero value across three resolutions, constraints that converge at the code's order while it grows, and a gauge-invariant measure—the throat's areal radii, or the marginally outer-trapped surface (MOTS) found spectrally on every output—growing with it. The production grid is a box of half-size $L = 128$ with three nested refinements on the throat, run at $N = 192/256/384$ base points (finest $\Delta x = 0.083/0.0625/0.042$); the evolutions reported in Sec. V ran on the $L = 64$, $N = 128$ grid of the static hold of Ref. [11] (same finest $\Delta x = 0.0625$), whose outer-boundary return sets their trust window at $t \approx 64$.

The evolutions form a ladder of five spins on the drainhole family (Table II). The static member is the reference for every rate; $v_e = 0.21$ lies inside the range of the $O(J^2)$ theory [17, 18], so the nonlinear rate there can be checked against the perturbative one; $v_e = 0.51$ is the main case; $v_e = 0.70$ and 0.90 form the fast end, beyond any perturbative result. Each rung is one resolution triplet.

Three checks validate the pipeline. (i) At $t = 0$ the Hamiltonian and momentum constraints converge at fourth order inside the throat region and are limited outside only by the tables' interpolation error (growing as $1/\Delta x^2$ and $1/\Delta x$ respectively), and the throat's ray radii

TABLE II. The evolution ladder: the drainhole family (a) at five spins, in code units ($\eta_0 = 2$, static mass $M = 1$). $M = \mu_+$ is the mass of the $\eta \to +\infty$ end; $J/M^2 > 1$ exceeds the Kerr bound. The $L = 64$ column gives the single-resolution holds of Sec. V, trusted to $t = 64$ ($t = 40$ at $v_e = 0.70$, where the far-end puncture limits the run), with τ the e-fold time of the equatorial throat radius. Each rung runs as a triplet on the $L = 128$ box at $N = 192/256/384$ (finest $\Delta x = 0.083/0.0625/0.042$); the static $N = 256$ member is the $L = 128$ hold of Ref. [11]. The last column gives the triplets' state at this draft and will hold their converged τ.

v_e	J	M	J/M^2	R_e	role	$L = 64$, $N = 128$ hold	$L = 128$ triplet
0	0	1	0	3.890	reference	collapse, $\tau = 5.8$; horizon at $t \approx 61$	$N = 192$, 384: *queued*
0.210	1.000	1.112	0.809	4.060	$O(J^2)$ check	collapse, $\tau = 6.8$	*queued*
0.513	4.111	1.860	1.188	5.254	main case	MOTS radius -0.61% at $t = 64$; no horizon	$N = 256$: *running*
0.699	12.17	3.311	1.110	7.773	fast	R_e unchanged to 10^{-4} for $t \le 40$	*queued*
0.900	138.2	11.68	1.013	23.82	fast	—	*queued*

converge to the solver's. We check this on the drainhole at $v_e = 0.51$ on three grids (ray radii within 5×10^{-5} at $N = 192$, $L = 64$) and, across the spin range, on the symmetric family at seven spins $v_e = 0.24$–0.99 with three resolutions each (within 7×10^{-5} at $v_e = 0.24$ to 9×10^{-4} at 0.99, $N = 256$). (ii) On the same symmetric set, the 3D ADM mass integral evaluated on the grid equals the same integral on the exact table metric to $\le 10^{-12}$, and its Richardson extrapolation meets the solver's mass to 2×10^{-5}–7×10^{-4} (at worst 0.7% relative, at $v_e = 0.99$); the ADM angular momentum matches to between 2×10^{-3} ($v_e = 0.24$) and 2×10^{-5} (0.99). (iii) Dynamically, the static drainhole run through this pipeline is a numerical twin of the production hold of Ref. [11] over the full record: $t = 0$ bit-identical, Hamiltonian norm within 5×10^{-7} for all $t \le 100$, collapse milestones within one time unit, and a fitted e-fold time $\tau(\delta R) = 5.27$ against 5.12 there.

V. SLOW SPIN: ROTATION SLOWS THE COLLAPSE

Perturbative initial data fail first. Our first spinning holds used the $O(J^2)$ backgrounds of Ref. [18] at $J = 0.25$ and $J = 1$ on the drainhole family. Both throats inflated where the static throat collapses—the same unstable mode with the opposite sign ($\tau = 5.8$ and 5.6). But expressed as the equivalent radial kick of Ref. [11], the push was $\varepsilon \approx -3 \times 10^{-5}$ at $J = 0.25$ and -7×10^{-3} at $J = 1$: growth by a factor $\sim$230 for a factor 4 in J, i.e. $\sim J^4$—the size of the $O(J^2)$ data's own truncation error, which the independent Einstein residual puts at 10^{-2} at $J = 1$ (Sec. III). Second-order data seed the very instability under study at their own error level; nothing at slow spin can be concluded from them.

On exact data the throat collapses, more slowly the faster it spins. On the solver tables (the $L = 64$ holds of Table II) the verdict reverses. The $J = 1$ drainhole ($v_e = 0.210$) collapses on the static mode: the equatorial radius falls 1% by $t = 46.9$ and 10% by $t = 61.8$, against the static hold's $t = 44.1$ and 57.0—an e-fold time of 6.8 against 5.8 by the same measure. At $v_e = 0.513$ the collapse slows much further: the MOTS radius is down only 0.61% at $t = 64$, the end of the trust window, with no horizon formed—where the static throat had shrunk 2.5% by $t = 49$ and detached a horizon at $t \approx 61$. Constraints hold at the static run's level throughout, and in each case the unperturbed hold is seeded only by discretization error, so the ordering of rates is the ordering of $|\omega_I|$.

Against the perturbative prediction. The static rate itself checks the linear theory: $\tau(\delta R) = 5.27$ against $\tau \approx 5.1$ from the $C = 1$ static mode of Ref. [17] at $\eta_0 = 2$. For the spin trend, Ref. [17] predicts $\Delta\omega_I^{(2)} > 0$ for every asymmetry—the main unstable mode weakens with spin. The measured sequence $\tau = 5.8$ ($J = 0$) $\to 6.8$ ($v_e = 0.21$) $\to >$several tens ($v_e = 0.51$) is that prediction's sign, now established nonlinearly and, at $v_e = 0.51$, far beyond the formal reach of the $O(J^2)$ expansion.

VI. CONCLUSIONS

Three results stand. First, the spinning Ellis–Bronnikov parameter space is now mapped as data, not curves: both families, from slow rotation to $v_e \approx 0.99$, as solutions exact to the 10^{-10} level at slow spin and the 10^{-7} level near extremality (Table I), including the massive drainhole family for which nothing was published—and including the correction of a field equation, Eq. (12) of Ref. [15], that closed-form extremal Kerr convicts directly. The practical lesson generalizes: a solver gated only on the equations it imposes would have passed every test while solving the wrong system; the unimposed identities caught it. Second, the pipeline from these solutions to 3+1 evolution is validated at fourth order, with the static limit a numerical twin of the production run of Ref. [11], and $O(J^2)$ initial data are shown to be unusable for stability work: their truncation error seeds the instability itself. Third, the first nonlinear evolutions of rotating wormholes show the radial collapse surviving but slowing with spin—$\tau = 5.8 \to 6.8$ from $J = 0$ to $v_e = 0.21$, and an order-of-magnitude slower decay at

$v_e = 0.51$—the sign predicted at $O(J^2)$ [17, 18], now beyond perturbation theory. Whether the instability disappears at fast spin, as the perturbative mode crossing suggests, or continues as an oscillating complex pair, as in the charged analogue [20], is what the fast rungs of the ladder (Table II) test.

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All evolutions use `GRTeclyn` [26], the AMReX-based [27] GPU port of `GRChombo` [28].

Appendix A: Solver details and the anchor comparison

The unknowns f, ω are expanded in Chebyshev polynomials of $x = (2/\pi)\arctan(\eta/\eta_0) \in [-1, 1]$, which compactifies both ends, and in even cosine modes of θ; derivatives are spectral, and the nonlinear system (4)–(5) is solved by Newton iteration with a direct linear solve. At each Newton step, ν and the constant D^2 are recovered together from the overdetermined pair—the first-order constraint (6) and the D^2 identity—as one linear least-squares problem, which is what makes the acceptance gates of Sec. III meaningful: the reported D^2 spread, Smarr defect, flux equality and full-Einstein residual are all computed from equations that never entered the solve. Spin is walked with J held fixed in each solve: ω_∞ joins the unknowns and one extra row imposes the target J, since ω_∞ is not monotone along a family (Table III). Steps are taken in $\log J$, predicted by the secant through the last two solutions, with pseudo-arclength in (state, J) as the fallback should J itself fold; grids refine automatically from 48×12 when the acceptance gates demand it, to 162×42 at $v_e = 0.95$ on the drainhole family. On the symmetric family the mirror identities are satisfied at the 10^{-14} level, including $\omega(0) = \omega_\infty/2$, and the equatorial ergosurface law of Ref. [15] holds to 10^{-12} wherever the ergosurface sits at $\eta > 0$ ($v_e \geq 0.77$). The extremal-Kerr substitution of Sec. II A is kept in the pipeline as a symbolic regression test.

TABLE III. Scalar charge D of the anchor solutions ($\eta_0 = 1$, $\omega_\infty = 0.5$) against Ref. [15]. Along each fixed-γ family, ω_∞ rises, turns over and descends, so $\omega_\infty = 0.5$ is reached twice; the published values sit on the post-turnover branch. Our values are grid-converged to the digits shown; the $\gamma = 2$ family continued to $J = 172$ confirms no third crossing exists. The two positive-γ offsets are the published values' finite-difference error, growing with γ; the pre-turnover branch is new.

γ	D (this work)	D [15]	D (pre-turnover, new)
-1	0.716368	0.71636	0.969194
0	0.552771	0.55277	0.986013
1	0.4314654	0.43155	1.008136
2	0.3375993	0.33984	1.048161

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